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Условие:

Первая задача о магазине. В магазине работают три отдела, каждый отдел обслуживает один продавец. Покупатель, зайдя в магазин, делает покупки в одном или нескольких произвольных 13 отделах, обходя их в произвольном (случайном) порядке. Если в выбранном отделе продавец не свободен, покупатель становится в очередь и ожидает, пока продавец не освободится. Создать многопроцессное приложение, моделирующее рабочий день магазина. Каждого покупателя и продавцов моделировать отдельными процессами. Размер очереди не оговаривается. Считается, что для данной задачи она не ограничена (но моделирование должно быть в разумных пределах).

1. Первая программа

Сценарий:

Магазин с 3 отделами. В каждом отделе работает один продавец. Покупатели заходят в магазин и посещают случайное количество отделов. Если продавец занят, покупатель ждет. Родительский процесс запускает всех покупателей и продавцов как дочерние процессы.

Процессы и взаимодействие:

- **Продавцы** – обслуживают покупателей, ожидают сигнал через семафоры.
- **Покупатели** – посещают отделы и сигнализируют продавцам через семафоры.
- **Семафоры** – неименованные POSIX семафоры в разделяемой памяти (`seller_free[i]` для отделов, `print_lock` для синхронного вывода).
- **Разделяемая память** – хранит семафоры и флаг `stop` для завершения работы.

Особенности реализации:

- `mmap` используется для общей памяти между процессами.
- Безопасный вывод через `print_lock`.
- `Ctrl+C` корректно завершает все процессы.

Запуск:

```
g++ -std=c++20 -pthread shop_day.cpp -o shop_day
```

```
./programm_1 4
```

- 4 – количество покупателей.
- `Ctrl+C` завершает все процессы корректно.

Пример вывода:

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_1$ ./programm_1 4
[Seller 0] started
[Seller 1] started
[Seller 2] started
    (Customer 0) entered shop
    (Customer 0) waiting for dept 0
[Seller 0] serving customer...
    (Customer 1) entered shop
    (Customer 1) waiting for dept 0
    (Customer 2) entered shop
    (Customer 2) waiting for dept 0
    (Customer 3) entered shop
    (Customer 3) waiting for dept 0
    (Customer 0) done in dept 0
    (Customer 0) leaving shop
    (Customer 1) done in dept 0
    (Customer 1) leaving shop
    (Customer 2) done in dept 0
    (Customer 2) leaving shop
[Seller 0] finished
[Seller 0] serving customer...
    (Customer 3) done in dept 0
    (Customer 3) leaving shop
[Seller 1] exiting
[Seller 2] exiting
[Seller 0] finished

[Seller 0] exiting
Main: finished
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_1$
```

Вывод при использовании сигнала:

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_1$ ./programm_1 10
[Seller 0] started
[Seller 1] started
[Seller 2] started
    (Customer 0) entered shop
    (Customer 0) waiting for dept 1
[Seller 1] serving customer...
    (Customer 1) entered shop
    (Customer 1) waiting for dept 1
    (Customer 2) entered shop
    (Customer 2) waiting for dept 1
^C    (Customer 2) done in dept 1
    (Customer 2) leaving shop
    (Customer 1) done in dept 1
    (Customer 1) leaving shop
[Seller 0] exiting
    (Customer 0) done in dept 1
[Seller 2] exiting
    (Customer 0) leaving shop
[Seller 1] finished
[Seller 1] exiting
    (Customer 3) entered shop
    (Customer 3) leaving shop
    (Customer 4) entered shop
    (Customer 4) leaving shop
    (Customer 5) entered shop
    (Customer 5) leaving shop
    (Customer 6) entered shop
    (Customer 6) leaving shop
    (Customer 7) entered shop
    (Customer 7) leaving shop
    (Customer 8) entered shop
    (Customer 8) leaving shop
    (Customer 9) entered shop
    (Customer 9) leaving shop
Main: finished
```

Очистка ресурсов: семафоры уничтожаются через `sem_destroy`, память освобождается `munmap`.

2. Вторая программа

Сценарий задачи:

Аналогично первой программе, но каждый процесс (покупатель/продавец) запускается независимо в разных консолях. Обмен данными через **именованные POSIX семафоры** и разделяемую память.

Особенности:

- Разделяемая память (shm_open) хранит флаг stop.
- Каждый продавец и покупатель открывает нужный семафор.
- Семафоры именованные (/sem_seller0, /sem_seller1, /sem_seller2, /sem_print).
- Вывод в свою консоль с синхронизацией через семафор SEM_PRINT.
- Завершение через Ctrl+C изменяет флаг stop в shared memory.

Запуск:

1. Инициализация магазина:

```
g++ -std=c++20 -pthread shop_init.cpp -o shop_init  
./shop_init
```

2. Запуск продавцов (по одному в каждом терминале):

```
g++ -std=c++20 -pthread seller.cpp -o seller  
./seller 0  
./seller 1  
./seller 2
```

3. Запуск покупателей:

```
g++ -std=c++20 -pthread customer.cpp -o customer  
./customer 0  
./customer 1  
./customer 2
```

4. Ctrl+C завершает все процессы корректно (использовано в seller 0).

Пример вывода (консоли покупателей):

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_2$ ./customer 0
(Customer 0) entered shop
(Customer 0) waiting for dept 0
(Customer 0) done in dept 0
(Customer 0) leaving shop
```

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_2$
```

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_2$ ./customer 1
(Customer 1) entered shop
(Customer 1) waiting for dept 2
(Customer 1) done in dept 2
(Customer 1) leaving shop
```

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_2$
```

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_2$ ./customer 2
(Customer 2) entered shop
(Customer 2) waiting for dept 1
(Customer 2) done in dept 1
(Customer 2) waiting for dept 1
(Customer 2) done in dept 1
(Customer 2) leaving shop
```

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_2$
```

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_2$ ./customer 3
(Customer 3) entered shop
(Customer 3) waiting for dept 0
(Customer 3) done in dept 0
(Customer 3) waiting for dept 1
(Customer 3) done in dept 1
(Customer 3) waiting for dept 2
(Customer 3) done in dept 2
(Customer 3) leaving shop
```

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_2$
```

Пример вывода (консоли продавцов):

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_2$ ./seller 0
[Seller 0] started
[Seller 0] serving customer...
[Seller 0] finished
[Seller 0] serving customer...
[Seller 0] finished
[Seller 0] exiting

hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_2$ ./seller 1
[Seller 1] started
[Seller 1] serving customer...
[Seller 1] finished
[Seller 1] serving customer...
[Seller 1] finished
[Seller 1] serving customer...
[Seller 1] finished
[Seller 1] exiting

hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_2$ ./seller 2
[Seller 2] started
[Seller 2] serving customer...
[Seller 2] finished
[Seller 2] serving customer...
[Seller 2] finished
[Seller 2] exiting
```

Как только закончились покупатели, нужно закрыть магазин, то есть прожать Ctrl+C (делается у одного продавца, в нашем случае Seller 0, остальные же закончат работу автоматически):

```
^C
[Seller] Caught SIGINT, shutting down...
[Seller 0] exiting
```

3. Третья программа

Сценарий задачи:

Добавлен **наблюдатель** — отдельный процесс, который интегрирует и отображает информацию от всех процессов. Покупатели и продавцы отправляют сообщения через **именованный канал (FIFO)** наблюдателю.

Особенности:

- Каждый процесс выводит в свою консоль и пишет в FIFO.
- Наблюдатель читает из FIFO и отображает интегрированную информацию.
- Поддержка Ctrl+C: завершение процессов через сигнал SIGINT и изменение флага stop.
- Разделяемая память и семафоры используются для синхронизации вывода и моделирования занятости продавцов.

Запуск:

1. *Создание FIFO:*

```
mkfifo /tmp/shop_fifo
```

2. *Запуск наблюдателя (в отдельной консоле):*

```
g++ -std=c++20 -pthread observer.cpp -o observer
```

```
./observer
```

3. *Запуск продавцов и покупателей (как во второй программе):*

```
./seller 0
```

```
./seller 1
```

```
./seller 2
```

```
./customer 0
```

```
./customer 1
```

```
./customer 2
```

```
./customer 3
```

4. *Ctrl+C корректно завершает все процессы (работает и у наблюдателя).*

Пример вывода (консоли продавцов):

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_3$ ./seller 0
[Seller 0] started
[Seller 0] serving customer...
[Seller 0] finished
[Seller 0] serving customer...
[Seller 0] finished
[Seller 0] serving customer...
[Seller 0] finished
[Seller 0] serving customer...
[Seller 0] finished
[Seller 0] serving customer...
[Seller 0] finished
[Seller 0] exiting
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_3$
```

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_3$ ./seller 1
[Seller 1] started
[Seller 1] exiting
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_3$
```

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_3$ ./seller 2
[Seller 2] started
[Seller 2] serving customer...
[Seller 2] finished
[Seller 2] serving customer...
[Seller 2] finished
[Seller 2] serving customer...
[Seller 2] finished
[Seller 2] serving customer...
[Seller 2] finished
[Seller 2] exiting
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_3$
```


Пример вывода (консоли покупателей):

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_3$ ./customer 0
(Customer 0) entered shop
(Customer 0) waiting for dept 2
(Customer 0) done in dept 2
(Customer 0) leaving shop
```

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_3$
```

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_3$ ./customer 1
(Customer 1) entered shop
(Customer 1) waiting for dept 0
(Customer 1) done in dept 0
(Customer 1) waiting for dept 2
(Customer 1) done in dept 2
(Customer 1) leaving shop
```

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_3$
```

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_3$ ./customer 2
(Customer 2) entered shop
(Customer 2) waiting for dept 2
(Customer 2) done in dept 2
(Customer 2) waiting for dept 2
(Customer 2) done in dept 2
(Customer 2) waiting for dept 0
(Customer 2) done in dept 0
(Customer 2) leaving shop
```

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_3$
```

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_3$ ./customer 3
(Customer 3) entered shop
(Customer 3) waiting for dept 0
(Customer 3) done in dept 0
(Customer 3) waiting for dept 0
(Customer 3) done in dept 0
(Customer 3) waiting for dept 0
(Customer 3) done in dept 0
(Customer 3) leaving shop
```

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_3$
```

Пример вывода (консоль наблюдателя):

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_3$ ./observer
[Seller 0] started
[Seller 1] started
[Seller 2] started
    (Customer 0) entered shop
    (Customer 0) waiting for dept 2
[Seller 2] serving customer...
    (Customer 0) done in dept 2
    (Customer 0) leaving shop
[Seller 2] finished
    (Customer 1) entered shop
    (Customer 1) waiting for dept 0
[Seller 0] serving customer...
    (Customer 1) done in dept 0
    (Customer 1) waiting for dept 2
[Seller 2] serving customer...
[Seller 0] finished
    (Customer 1) done in dept 2
    (Customer 1) leaving shop
[Seller 2] finished
    (Customer 2) entered shop
    (Customer 2) waiting for dept 2
[Seller 2] serving customer...
    (Customer 2) done in dept 2
    (Customer 2) waiting for dept 2
[Seller 2] finished
[Seller 2] serving customer...
    (Customer 2) done in dept 2
    (Customer 2) waiting for dept 0
[Seller 0] serving customer...
[Seller 2] finished
    (Customer 2) done in dept 0
    (Customer 2) leaving shop
[Seller 0] finished
    (Customer 3) entered shop
    (Customer 3) waiting for dept 0
[Seller 0] serving customer...
    (Customer 3) done in dept 0
    (Customer 3) waiting for dept 0
[Seller 0] finished
[Seller 0] serving customer...
    (Customer 3) done in dept 0
    (Customer 3) waiting for dept 0
[Seller 0] finished
[Seller 0] serving customer...
    (Customer 3) done in dept 0
    (Customer 3) leaving shop
[Seller 0] finished
^C
[Observer] Caught SIGINT, shutting down...
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_3$
```

4. Четвертая программа

Сценарий задачи:

Добавлена поддержка **нескольких наблюдателей**, каждый выводит информацию в свою консоль. Все процессы пишут в все FIFO.

Особенности:

- FIFO создаются для каждого наблюдателя: /tmp/shop_fifo_0, /tmp/shop_fifo_1 и т.д.
- Каждый процесс пишет в все наблюдатели.
- Семафоры используются для синхронизации вывода и обслуживания покупателей.
- Завершение через Ctrl+C: изменение флага stop в разделяемой памяти.
- Разделяемая память и семафоры корректно удаляются после завершения.

Запуск:

1. *Инициализация магазина (создание shared memory и FIFO для наблюдателей):*

./shop_init

2. *Запуск наблюдателей(разные наблюдатели-разные консоли):*

./observer 0

./observer 1

3. *Запуск продавцов и покупателей:*

./seller 0

./seller 1

./seller 2

./customer 0

./customer 1

./customer 2

./customer 3

4. *Ctrl+C корректно завершает все процессы.*

Пример вывода (консоли продавцов):

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_4$ ./seller 0
[Seller 0] started
[Seller 0] serving customer...
[Seller 0] finished
[Seller 0] serving customer...
[Seller 0] finished
[Seller 0] serving customer...
[Seller 0] finished
[Seller 0] exiting
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_4$
```

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_4$ ./seller 1
[Seller 1] started
[Seller 1] serving customer...
[Seller 1] finished
[Seller 1] serving customer...
[Seller 1] finished
[Seller 1] serving customer...
[Seller 1] finished
[Seller 1] exiting
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_4$
```

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_4$ ./seller 2
[Seller 2] started
[Seller 2] serving customer...
[Seller 2] finished
[Seller 2] exiting
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_4$
```

Пример вывода (консоли покупателей):

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_4$ ./customer 0
(Customer 0) entered shop
(Customer 0) waiting for dept 1
(Customer 0) done in dept 1
(Customer 0) leaving shop
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_4$

hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_4$ ./customer 1
(Customer 1) entered shop
(Customer 1) waiting for dept 0
(Customer 1) done in dept 0
(Customer 1) waiting for dept 0
(Customer 1) done in dept 0
(Customer 1) leaving shop
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_4$ █

hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_4$ ./customer 2
(Customer 2) entered shop
(Customer 2) waiting for dept 1
(Customer 2) done in dept 1
(Customer 2) waiting for dept 1
(Customer 2) done in dept 1
(Customer 2) leaving shop
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_4$

hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_4$ ./customer 3
(Customer 3) entered shop
(Customer 3) waiting for dept 0
(Customer 3) done in dept 0
(Customer 3) waiting for dept 2
(Customer 3) done in dept 2
(Customer 3) leaving shop
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_4$
```

Пример вывода (консоли наблюдателей):

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_4$ ./observer 0
[Seller 0] started
[Seller 1] started
[Seller 2] started
  (Customer 0) entered shop
  (Customer 0) waiting for dept 1
[Seller 1] serving customer...
  (Customer 0) done in dept 1
  (Customer 0) leaving shop
[Seller 1] finished
  (Customer 1) entered shop
  (Customer 1) waiting for dept 0
[Seller 0] serving customer...
[Seller 0] finished
  (Customer 1) done in dept 0
  (Customer 1) waiting for dept 0
[Seller 0] serving customer...
  (Customer 2) entered shop
  (Customer 2) waiting for dept 1
[Seller 1] serving customer...
[Seller 0] finished
  (Customer 1) done in dept 0
  (Customer 1) leaving shop
  (Customer 2) done in dept 1
  (Customer 2) waiting for dept 1
  (Customer 2) done in dept 1
  (Customer 2) leaving shop
[Seller 1] finished
[Seller 1] serving customer...
[Seller 1] finished
  (Customer 3) entered shop
  (Customer 3) waiting for dept 0
[Seller 0] serving customer...
  (Customer 3) done in dept 0
  (Customer 3) waiting for dept 2
[Seller 2] serving customer...
[Seller 0] finished
  (Customer 3) done in dept 2
  (Customer 3) leaving shop
[Seller 2] finished
^C
[Observer] Caught SIGINT, shutting down...
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб/CLionProjects/IHW33/programm_4$
```

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб6/CLionProjects/IHW33/programm_4$ ./observer 1
```

```
[Seller 0] started
[Seller 1] started
[Seller 2] started
  (Customer 0) entered shop
  (Customer 0) waiting for dept 1
[Seller 1] serving customer...
  (Customer 0) done in dept 1
  (Customer 0) leaving shop
[Seller 1] finished
  (Customer 1) entered shop
  (Customer 1) waiting for dept 0
[Seller 0] serving customer...
[Seller 0] finished
  (Customer 1) done in dept 0
  (Customer 1) waiting for dept 0
[Seller 0] serving customer...
  (Customer 2) entered shop
  (Customer 2) waiting for dept 1
[Seller 1] serving customer...
[Seller 0] finished
  (Customer 1) done in dept 0
  (Customer 1) leaving shop
  (Customer 2) done in dept 1
  (Customer 2) waiting for dept 1
  (Customer 2) done in dept 1
```

```
  (Customer 2) leaving shop
[Seller 1] finished
[Seller 1] serving customer...
[Seller 1] finished
  (Customer 3) entered shop
  (Customer 3) waiting for dept 0
[Seller 0] serving customer...
  (Customer 3) done in dept 0
  (Customer 3) waiting for dept 2
[Seller 2] serving customer...
[Seller 0] finished
  (Customer 3) done in dept 2
  (Customer 3) leaving shop
[Seller 2] finished
```

```
hleb@LAPTOP-EFD83A08:/mnt/c/Users/Глеб6/CLionProjects/IHW33/programm_4$
```